Research topic: The Evolution of Automated Fake News Detection

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Abstract:

The proliferation of misinformation on digital platforms has escalated into a significant societal challenge, threatening public trust and democratic stability. This research project investigates the technological evolution of automated fake news detection, tracing its trajectory from early computational linguistics to the current era of Generative Artificial Intelligence.

In its initial stages, detection relied heavily on **Traditional Machine Learning** (e.g., SVM, Naïve Bayes, and Logistic Regression). These models focused on "hand-crafted" features, such as n-grams, punctuation density, and readability scores, to identify the sensationalist style often associated with clickbait. However, as misinformation evolved to mimic professional journalism more closely, these methods lost effectiveness.

The research then examines the shift toward **Deep Learning** architectures, specifically Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, which allowed for better capturing of sequential context. This was further revolutionized by **Transformer-based models** (like BERT and RoBERTa), which introduced the "attention mechanism," enabling the detection of subtle semantic nuances that previous models missed.

The final phase of this project explores the current state-of-the-art: **Large Language Models (LLMs)** and **Hybrid Graph Neural Networks (GNNs)**. Unlike earlier classifiers, LLMs provide "explainable" detection by reasoning through evidence, while GNNs analyze the propagation patterns—how the news spreads—rather than just the text itself. By benchmarking these methods against standard datasets (e.g., LIAR, WELFake), this research aims to identify the trade-offs between computational efficiency and detection accuracy in a real-time information environment.

References (at least 3):
Xu, C., et al. (2026). *LiveFact: A Dynamic, Time-Aware Benchmark for LLM-Driven Fake News Detection.* arXiv preprint arXiv:2604.04815. (This focuses on the most recent LLM reasoning capabilities).
Kumar, A., et al. (2025). *Fake News Detection: Classifying Articles as Real or Fake Using NLP.* Proceedings of the 2025 International Conference on Science and Technology. (Covers the transition from traditional ML to Deep Learning).
Zhou, X., & Zafarani, R. (2021). *A Survey of Fake News: Fundamental Theories, Propagation, and Detection.* ACM Computing Surveys. (A foundational text for understanding the "older" methods and the sociology of fake news).
Jin, C., et al. (2025). *Fake News Detection Using Hybrid Transformer-Based Models.* International Journal of Science and Advanced Technology. (Explains the combination of RoBERTa and Graph Neural Networks).